EXPERIMENT 5

Determination and plotting of Electric potential difference between any two points in free space medium separated by a distance R.

Aim:

To write program for Electric potential difference between two points in free space medium using SCILAB.

Software required:

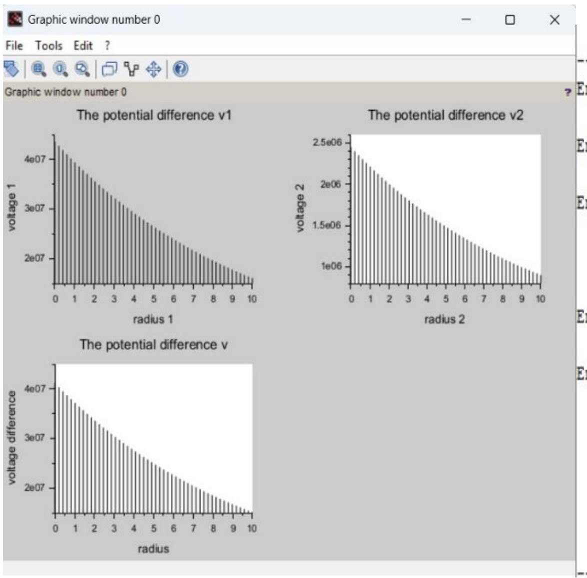
- SCILAB version 6.0.1

Code:

exp 2 case 1.sci   exp 3.sci   exp 3case 1.sci

```
clc;
clear;
q=input("Enter the q value:");
e=8.854*10^-12;
pi=3.14;

ax=input('Enter the value of ax ');
ay=input('Enter the value of ay ');
r1=sqrt(ax^2+ay^2);
v1=q/(4*pi*e*r1);
disp('The value of v1',[v1]);

bx=input('Enter the value of bx ');
by=input('Enter the value of by ');
r2=sqrt(bx^2+by^2);
v2=q/(4*pi*e*r2);
disp('The value of v2',[v2]);

r=r1-r2;
v=v1-v2;
disp('The electric potential difference is',[v]);
r1 = linspace(0, 10, 50);
r2= linspace(0, 10, 50);
r= linspace (0, 10, 50);
y = v1*exp(-0.1 * r1);
z = v2*exp(-0.1 * r2);
x = v*exp(-0.1 * r);

figure();
subplot(2, 2, 1);
plot2d3(r1,y);
xlabel("radius 1");
ylabel("voltage 1");
title('The potential difference v1');
subplot(2, 2, 2);
plot2d3(r2,z);
xlabel("radius 2");
ylabel("voltage 2");
title('The potential difference v2');
subplot(2, 2, 3);
plot2d3(r,x);
xlabel("radius");
ylabel("voltage difference");
title('The potential difference v');
```

Output:




```
--> exec('C:\Users\jasiy\OneDrive\Documents\exp 5.sci', -1)
Enter the q value:2*10^-2

Enter the value of ax=1

Enter the value of ay=4

  "The value of v1"
   43619068.
Enter the value of bx=36

Enter the value of by=64

  "The value of v2"
   2449209.3
  "The electric potential difference is"
   41169858.

-->
```

Theoretical Calculation:

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Output :

Enter the q value = 2×10^{-2}

Enter the value ax = 1

Enter the value ay = 4

Enter the value of bx = 36

Enter the value of by = 64

Practical output :

"The value of v_1"

43619068.

"The value of v_2"

2449209.3

"The electric potential difference is"

41169858.

Theoretical calculation :

Distances of A: $r_1 = \sqrt{1^2+4^2} = \sqrt{17} = 4.1231m$

Distance of B : $r_2 = \sqrt{36^2+64^2} = \sqrt{1296+4096} = \sqrt{5392} = 73.43$

Electrical potential at A

$$v_1 = \frac{1}{4\pi\varepsilon_0}\frac{q}{r_1} = \frac{9\times10^9 \times 2\times10^{-2}}{4.1231} = 4.366\times10^7 v$$

$v_1 = 43660000 v$

Electrical potential at B

$$v_2 = \frac{9\times10^9 \times 2\times10^{-2}}{73.4302} = 2.451\times10^6 v$$

$v_2 = 2451000 v$

Potential Difference

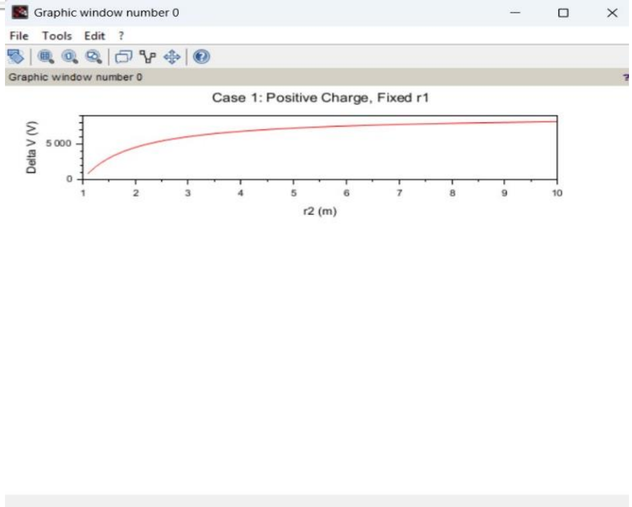
$\Delta v = v_1 - v_2 = 4.366\times10^7 - 2.451\times10^6$

$\Delta v = 4.121\times10^7 v$ or $41210000 v$

Case 1: A positive charge with fixed r1=1 m r2 varying.

Code: **output:**

```
// Constants
epsilon0 = 8.854e-12; // Permittivity of free space
k = 1 / (4 * %pi * epsilon0); // Coulomb's constant
q = 1e-6; // Charge in Coulombs (1 µC)

// Case 1: Positive charge, r1 = 1 m, r2 varies
r1_case1 = 1; // meters
r2_case1 = linspace(1.1, 10, 100); // meters
deltaV_case1 = k * q * (1 / r1_case1 - 1 ./ r2_case1);



// Plotting
clf();
subplot(3, 1, 1);
plot(r2_case1, deltaV_case1, "r");
xlabel("r2 (m)");
ylabel("Delta V (V)");
title("Case 1: Positive Charge, Fixed r1");

```

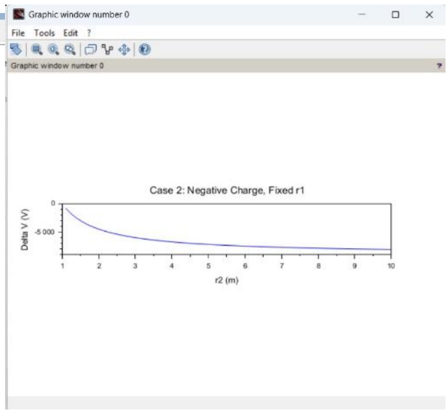


Case 2: A negative charge with r1r_1r1 fixed and r2r_2r2 varying.

Code: **output:**

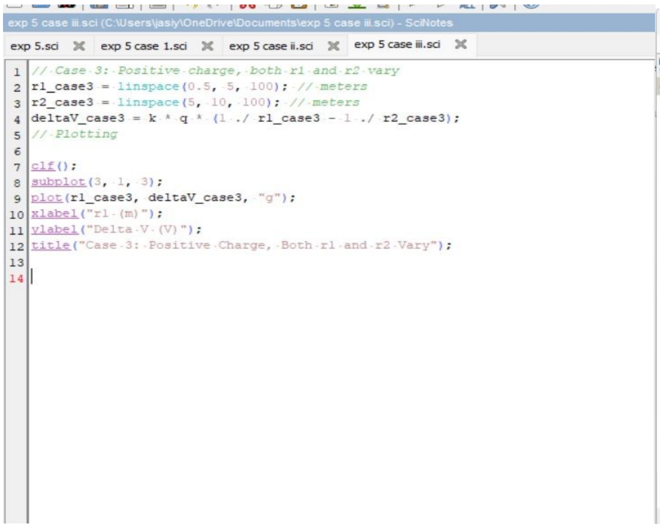
exp 5.sci | exp 5 case 1.sci | exp 5 case ii.sci

```
// Case 2: Negative charge, r1 fixed, r2 varies
q_case2 = -1e-6; // -1 µC
r1_case2 = 1; // meters
r2_case2 = linspace(1.1, 10, 100); // meters
deltaV_case2 = k * q_case2 * (1 / r1_case2 - 1 ./ r2_case2);
// Plotting

clf();
subplot(3, 1, 2);
plot(r2_case2, deltaV_case2, "b");
xlabel("r2 (m)");
ylabel("Delta V (V)");
title("Case 2: Negative Charge, Fixed r1");

```

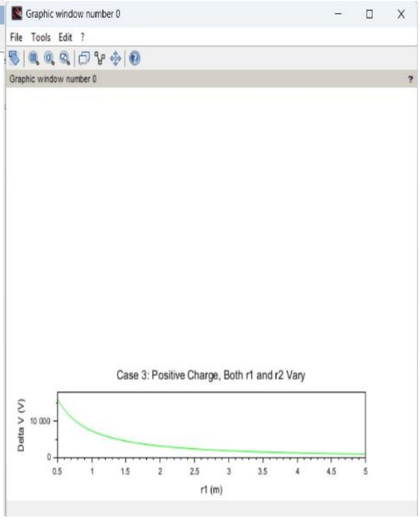


Case 3: Both points r1r_1r1 and r2r_2r2 vary with a positive charge.

Code: **output:**



```
// Case 3: Positive charge, both r1 and r2 vary
r1_case3 = linspace(0.5, 5, 100); // meters
r2_case3 = linspace(5, 10, 100); // meters
deltaV_case3 = k * q * (1 ./ r1_case3 - 1 ./ r2_case3);
// Plotting

clf();
subplot(3, 1, 3);
plot(r1_case3, deltaV_case3, "g");
xlabel("r1 (m)");
ylabel("Delta V (V)");
title("Case 3: Positive Charge, Both r1 and r2 Vary");

```



Result :

Thus the program was verified for Electric potential difference between two points in free space medium using SCILAB was successfully.